

Patrick PRELL

Robotics & Embedded Systems Engineer

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I'm a mechatronics engineer drawn to where mathematics meets a physical system. Taking a problem from the whiteboard through to hardware that works. Across eight years I've built state estimation and computer vision for survey UAVs at CORDEL, safety-critical medical firmware at ResusRight, and RF direction-finding at DroneShield. I'm as comfortable in the algorithms as in the embedded C++/Python that runs them.

SKILLS

Control & Estimation	Extended Kalman Filter, Bayesian sensor fusion , full-state feedback, observer design, fixed-rate discrete-time control loops, MATLAB
Perception & Vision	OpenCV (C++) , optical flow, 3D reconstruction from 2D imagery, machine learning for point-cloud refinement, camera calibration, ground-truth evaluation
Robotics & Autonomy	ROS 2 , ArduPilot/PX4 guidance & control, mission planning, SITL simulation to hardware , GNSS/RTK navigation, localisation and pose estimation
Coordinate Frames & Calibration	Sensor-frame transforms (IMU / RTK GNSS / LiDAR) , multi-channel RF calibration, amplitude/phase correction, repeatable unit-to-unit accuracy
Algorithms & Numerics	Turning mathematical models into real-time implementations , numerical methods, model-vs-hardware debugging, research-grade methods to maintainable production code
Software & Embedded	C++, Python (numpy/scipy), Linux , Rust, bare-metal C/C++, STM32 / NXP / Silicon Labs / ESP32, Qt, gRPC, POSIX shell
Hardware & Delivery	PCB design (KiCad, Eagle), 3D CAD (PTC Creo, Fusion 360), prototype to production; led small teams to on-time delivery, mentoring and university tutoring

PROFESSIONAL EXPERIENCE

Current Nov 2023	Embedded Software Engineer DroneShield - Australia ➤ Delivered the conducted calibration behind the angle-of-arrival estimation for a four-antenna direction-finding system (400 MHz - 7 GHz). Designed simultaneous multi-channel injection through a characterised splitter, producing a per-channel correction table that removes RSSI imbalance compensating the bearing angle accuracy. ➤ Cut analog front-end calibration from over 12 hours to 40 minutes. A 20x improvement that removed the single largest bottleneck in production. Required a full rearchitecture and coordination across production, hardware and test engineering. ➤ Built Python (Dash/Plotly) analysis tooling for frequency sweeps and calibration results. Stitched S2P measurements across 11 filter banks into a single validated front-end response. ➤ Led the cleanup and modernisation of our C++ Qt display codebase; enforcing clang-format and lint rules, improving readability and cutting the bug count. Embedded C++ Python Rust RF Calibration Direction Finding FPGA/RFSoc Qt gRPC Embedded Linux numpy/scipy Dash/Plotly POSIX Shell Jira GitLab
May 2025 Sep 2019	Co-founder & CTO Borne Clothing - Australia ➤ Co-founded and grew a social-enterprise clothing business to a successful exit. Donated \$30,000+ to the Against Malaria Foundation, resulting in 5,000 families protected from mosquito-borne disease for three years. Business Operations Web Development Lean Growth
Sep 2023 May 2022	Embedded Systems Engineer ResusRight - Australia ➤ Led firmware development of a manufacturing test jig, managing a small team and setting timelines that delivered on schedule and unblocked board production. ➤ Architected a Qt application and BLE stack on an embedded Linux SOM for "Nemo", a clinical resuscitation monitor (NXP and Silicon Labs MCUs, BLE and serial stacks). ➤ Worked to IEC 62304, IEC 60601 and ISO 13485. Safety-critical, traceable firmware in a startup moving at pace. Embedded C/C++ Embedded Linux Qt NXP µC Silicon Labs µC Espressif µC Bluetooth LE IEC62304 IEC60601 ISO13485 Python GitHub

May 2022	Mechatronic Engineer CORDEL (formerly Airsight) - Australia	> Implemented an Extended Kalman Filter fusing IMU and RTK GNSS for pose estimation and localisation of a survey platform in unstructured outdoor environments; the resulting pose estimate registered the LiDAR scans into a coherent 3D point cloud. Handled the coordinate-frame transforms between sensors, the Bayesian filtering and linear algebra, and the measurement dropout and noise of real field data; reconciling the filter model with how the hardware actually behaved. > Built a C++/OpenCV perception pipeline that recovered 3D structure and point clouds from the optical flow of a monocular GoPro feed. Spent a year in New York, working largely unsupervised, developing this on 2D camera footage from a moving train. > Worked hands-on with ArduPilot and PX4 flight stacks across the company's survey UAV fleet. Airframe bring-up, flight controller configuration and tuning, mission planning and field flight operations.
Apr 2018		

ArduPilot
PX4
UAV Flight Operations
GNSS/RTK
Mission Planning
C++
Python
OpenCV
Extended Kalman Filter
Sensor Fusion
LiDAR
RTK GNSS
Point Clouds
Embedded C
AVR Assembly
RTOS
KiCad
Rust
Jira
Bitbucket

PROJECTS

Optical-flow obstacle avoidance for a fixed-wing UAV

2019

B.Eng. (Hons) thesis – University of Newcastle, supervised by Dr Chris Renton

github.com/Preilly95/Final-Year-Project

Vision-only (single-camera) obstacle avoidance that steers on the divergence of the optical-flow field. Derived the flow kinematics from first principles on a unit view-sphere (6-DOF camera motion, rotation matrices across camera/body/world frames, inverse-depth parameterisation) and posed heading selection as a divergence-versus-goal cost-function optimisation. Built a full camera and lens-distortion calibration pipeline (undistortion in C via a MATLAB MEX) and validated it in a Unity physics simulation against ground-truth depth.

Optical Flow
 Computer Vision
 Kinematics
 Coordinate Transforms
 Camera Calibration
 Optimisation
 C++/OpenCV
 MATLAB
 C (MEX)
 Unity

Farm inspection drone in ROS 2 / PX4 SITL

2026

github.com/Preilly95/paddock

Autonomous ROS 2 (rclcpp) inspection stack for a VTOL platform. Guidance and control for autonomous take-off/land and fixed-wing cruise between paddocks, with the EKF-based state estimate and mission logic developed and tested in Gazebo/PX4 SITL simulation before flying on hardware. Missions gated on live flight conditions; telemetry feeds a homestead C2 dashboard for water level and asset monitoring.

ROS 2
 rclcpp
 PX4
 uXRCE-DDS
 Gazebo
 EKF
 Guidance & Control
 Simulation-to-Hardware
 C++
 Python

Two-wheeled balancing robot

2017

github.com/Preilly95/MCHA3K

Took the full loop from mathematics to hardware: derived and linearised the plant model, designed a Kalman observer and full-state feedback controller in MATLAB, then implemented the discrete-time control law at 50 Hz in embedded C on an 8-bit ATmega32 with no FPU. End-to-end build including the PCB.

Full-State Feedback
 Kalman Observer
 Real-Time Control
 Plant Modelling
 Embedded C
 PCB Design
 MATLAB
 Eagle

EDUCATION

Jul 2019	B.Eng. Mechatronics (Honours) University of Newcastle	
Feb 2015		
Feb 2022	Vision-Based Navigation Industry Short Course - Dr Chris Renton	
Jan 2022		
Feb 2023	QMS ISO 13485 Training Industry Short Course - SeerPharma	
Feb 2023		

REFERENCES

Available on request.